

The main part of this manuscript aims for two somewhat unrelated sets of contributions. The main set, which is discussed in the abstract, consists of a replication and extension of existing work on musical taste schemas using a slightly modified version of Boutyline's (2017) CCA. Because existing work on musical taste schemas has been focused on the same increasingly dated 1993 GSS musical tastes module, the author's replication with novel data is highly welcome, and the theoretical interpretation of this replication in terms of "cultural malleability" (Vaisey and Lizardo 2016) is also interesting and well executed. My only substantive concerns with these analyses involve the author's interpretation of one of the results. I believe this is something the author could easily address, and offer some suggestions towards an alternate interpretation that I believe fits the data better. I similarly point out places where the theoretical exposition could be clarified.

This main part is followed by a two of behavioral validation tests based on a methodological extension to CCA. I think a behavioral validation is a great contribution, and the extension is also promising. Again, my concerns with this part focus mainly on explanation and interpretation, which I thought was currently quite opaque. Below I untangle the logic of this section by recourse to the literature from which the author draws the methods and design of the validation analyses. I find that the first validation test actually makes a good deal of sense, while the second one is highly dubious. However, the paper makes sufficient contributions without the second study, and it is unrelated to the remaining analyses. My primary recommendation regarding the second validation is to simply remove it.

Below I detail these concerns. But first, I want to note that this review is clearly longer than the Socius format suggests. I want to apologize for this and make clear that this review's length should not be taken as a negative judgement on the manuscript under consideration. My up-and-down recommendation for the manuscript is unambiguously "up." To make the review more closely resemble the Socius format, I divide each of my comments into a part labelled ISSUE, where I describe my concern with the manuscript, and a part labelled SUGGESTION, where I propose ways the author could address the issue without incurring an excessive amount of extra work.

1. ISSUE: attribution of prior work. The design of both of the author's validation studies, as well as the "spreading activation" model of correlation networks s/he uses to describe them (4.3 – 4.3.3), are both closely adapted from Robinaugh et al. (2016). While the author cites Robinaugh frequently, and correctly indicates that the EI centrality measure comes from Robinaugh et al. (2016), s/he does not make it clear that the logic of sections (4.3 – 4.3.3) is almost entirely adapted from Robinaugh as well. Similarly, the terminology and theoretical framework of the author's main analyses are directly adapted from Boutyline (2017). Again, Boutyline (2017) is cited frequently in the paper, and the author makes clear that this is the origin of CCA and of the results s/he is replicating. But s/he does not make it clear that the theoretical framework and terminology described in (1.3) is also directly borrowed from Boutyline.

SUGGESTION: Much of this manuscript is adapted from either Boutyline (2017) or Robinaugh et al. (2016). This is obviously not a problem, and the combination of the two makes for an interesting and compelling contribution. However, the author needs to indicate the scope of these borrowings more clearly, especially in sections (4.3) and (1.3). The author should also point out that the reader can consult these source texts for more details on the models and methods.

2. ISSUE: the name and description of the Art/Pop/Folk schema on pp. 7-8 are both confusing and do not seem to fit the data well. At first, I did not intuitively get what kind of partitioning of the genre space was implied by this name. This became a bit clearer to me when I realized that by "Art" the author seems

to have meant “Highbrow” (which is the term generally used in the literature, e.g., van Eijk [2001] and Goldberg [2011].) But looking at Figure 2(e), neither Art nor Highbrow seem like a particularly accurate description of the results: the part of the network the author calls “Art” consists of a tightly correlated cluster of opera, classical, easy listening, oldies, musicals and big band (and more peripherally oldies and jazz). This set of genres is less like “highbrow” (or “art”) and more like “things that older folks like.” This is a lot like the Traditional cluster in Contemporary/Traditional, except the four genres popular in rural areas (Folk, Bluegrass, Gospel, Country) are split off from the rest of the Traditional tastes into their own cluster. The Pop cluster (which is a somewhat misleading name here because Pop music is not a core member of it) is in turn quite similar to the Contemporary cluster. The overall logic thus seems to be roughly Contemporary / Traditional-Urban / Traditional-Rural—so, as a first stab at a title, maybe something like “Contemporary / Soothing / Folksy”?

SUGGESTION: The author should consider renaming this schema. Since s/he has access to more data than me, s/he may disagree, or may find that some third description provides a more apt summary. But in either case, the author should rewrite this paragraph: it currently consists of too much lingo and no common-language intuitive interpretation of the results. Doing so may also clarify to the author which schema name to choose.

3. ISSUE: The author also does not do a great job of explaining either the intuitions behind Robinaugh et al.’s “expected influence” measure EI (which is simply weighted degree centrality as defined by Barrat et al. [2004] *PNAS* paper, which the author should also cite.) S/he introduces it with virtually no context, stating that “This index produces a rank of genres for each schematic class such that those with strong positive correlations with other genres are expected to be the most influential [...],” without ever explaining what “influential” means in this context. S/he also does not make clear that this meaning of “influential” is based specifically on the “spreading activation” view of correlation networks. The author’s explanation of the spreading activation model could also be better: currently s/he provides a definition of “spreading activation” which itself uses the term “spread” (without providing a definition,) and is thus circular. It would be clearer if the author noted that s/he is borrowing this view of correlation networks from Robinaugh et al. (2016), and followed that paper’s careful and clear exposition. Robinaugh’s work sits in the literature on correlation networks in clinical psychology, where this model is in common use. This literature is specifically focuses on correlation networks between mutually reinforcing symptoms, where a positive tie between A and B means that an increase in A is expected to probabilistically cause an increase in B and vice versa. In this context, the fit of the spreading activation framework is straightforward: if B is also positively tied to C, which is negatively tied to D, then that increase in B would in turn probabilistically lead to an increase in C, which would probabilistically lead to a decrease in D, etc. This explains what is meant by “influence” above.

The author uses his or her a rhetorical gloss of the spreading activation model to argue that genres with the highest EI should have the highest influence on the state of overall network, because they by definition have the strongest ties to other genres. But this does not immediately follow, because we also need to consider activation spreading on to second, third, etc., nodes. Instead, this conjecture is a central subject of Robinaugh et al.’s (2016) work, who used both simulations and empirical validation to establish it to be true. This model-derived conjecture and empirical validation are the intuitive reason Robinaugh et al. called their centrality measure “expected influence.” (I explain the design of Robinaugh’s validation studies below, because they make the logic this approach much clearer.)

* SUGGESTION: The EI measure, spreading activation model, and conjecture of overall network influence all come from Robinaugh et al. (2016). The author should attribute this authorship properly, and use Robinaugh’s work to explain how these pieces fit together, much as I do above.

4. ISSUE: The validation in section 4.3.2 is difficult to follow: its logic is complex, and the author devotes only a single page to describing the theory, method, and results of this analysis. Fundamentally, the validation in 4.3.2 makes sense—but the author needs work on the exposition for this to come through. The author’s validation strategy here is also adapted from Robinaugh et al. That paper contains a simulation and an empirical validation. In the simulation, Robinaugh et al. randomly generated correlation networks between disorder symptoms. They then simulated spreading activation by setting an activation value of random nodes to 0 or 1, and propagating that change of activation throughout the rest of the network. They used this to show that the EI of a node correctly predicts how much effect its change of activation would have on the rest of the network. They then (B) validated EI empirically with panel clinical data on patients measured at time points $t1$ and $t2$. For each time point, they first calculated the total count of each patient’s symptoms ($symptoms_t1$ and $symptoms_t2$). They then calculated how much each patient’s condition changed by taking a difference between the two, $change_t1_t2 = symptoms_t2 - symptoms_t1$. They then calculated how well each of the patient’s symptoms j at $t1$ (j_t1) could predict the change in the patient’s condition, $predictiveness_j = cor(j_t1, change_t1_t2)$. Finally, to validate EI, they calculated how much a symptom’s centrality EIj tracks its predictiveness, $Cj = cor(EIj, predictiveness_j)$, and demonstrated that $Cj > 0$.

The author of the present paper basically describes the spreading activation model behind (A), and then adapts the design of validation (B) to his or her data. This this data is cross-sectional, s/he cannot calculate $change_t1_t2$. In its place, s/he uses a behavioral measure of univorousness/omnivorousness, which is simply the total count of genres a respondent reports actually listening, $listens_t1$ (not to be confused with the genres the respondent simply reports liking, which is the data behind the correlation networks.) This leads the author to replace $predictiveness_j$ with $importance_j = cor(j_t1, listens_t1)$. The validation results the author reports in figure 5 show that a node’s centrality EIj tracks this behavioral measure of importance, i.e., $cor(EIj, importance_j) > 0$.

The author’s own description of this validation study is quite a bit more opaque than the description I provide above, which I gleaned mostly from Robinaugh. This is partly because s/he does not first discuss Robinaugh’s simulation study, which would provide context, and partly because s/he does not use any formulas or mathematical notation. It is also partly because s/he often forgoes simple names in favor of lengthy technical descriptions—e.g., the section concludes with: “This constitutes strong evidence for the [external] validity of the correlation networks extracted from each taste schema and the spreading activation interpretation of the link between the resulting schemas and cultural choice behavior.” This statement could be made clearer if “the correlation networks extracted from each taste schema” was replaced by “CCA results,” and “the spreading activation interpretation of the link between the resulting schemas and cultural choice behavior” was replaced with “predictions of the spreading activation model.”

* SUGGESTION: The author could do a number of things to clarify his or her account. First, much like the spreading activation model and the EI measure, this validation procedure is clearer if described in the context of Robinaugh’s et al.’s (2016). I think the logic of Robinaugh’s simulation study is especially helpful for understanding what’s going on here. Second, the author should include variable names and formulas like I did above, and break up complex operations into separate steps with names assigned to intermediate results (this is difficult material and so it requires a bit more room to become clear). Finally s/he should add further descriptions; simplify the sentence structure; and remove excess lingo.

5. ISSUE: The analyses in 4.3.3 are extremely hard to understand. This is at least in part because the author uses a total of three paragraphs to introduce a new theoretical concept, describe the method behind

a rather complex empirical analysis, describe its results (found across Figures 6, 7a, and 7b), and interpret the meaning of these results as both a validation study and an empirical statement about omnivorousness. The concrete stumbling block I hit while reading this section concerned its focal concept of “omnivore profiles,” which it consists of genres that omnivores consume more than others in that SCA assigned to their schematic class. Since classes are abstract assignments wedded to the logic of SCA, it is very hard for me to wrap my mind around what this measure means empirically, and to have any clear intuitions about how it should behave. I can see this measure as potentially one useful step within a detailed analysis of a schematic class, but making it useful would take a lot of careful handling and interpretation. That is not what the author presently does in this section. It also does not help that Figure 6 (which depicts these omnivore profiles) lacks a caption or axis labels, which are also not described in the text. Key information, like the number of respondents who fall into the “omnivorous” group within each schema, is also not provided. Additionally, the same problems with clarity I note for 4.3.2 also apply here. As a consequence, I found 4.3.3 virtually impenetrable.

* SUGGESTION: I think it would take a substantial amount of work to turn 4.3.3 into a clear and convincing analysis. If the author chooses to do so, I recommend they separate this section into separate parts for the theory, methods, and results reported here. My recommendations regarding language and formulas from 4.3.2 also apply. But I also think this section is not doing much work for the paper: though it sits under the heading of “validation,” it uses the same data and mostly the same measures as 4.3.2., and does not appear to provide any additional evidence towards the validity of CCA results or the EI measure. It also has nothing to do with the cultural durability thesis the author discusses in the main analyses. Thus, my primary recommendation is that the author simply remove this section in its entirety, as the manuscript makes more than enough contributions without it.

6. ISSUE: In section (1.3) the theoretical model (and terminology) described by the author are from Boutyline (2017). However, the author also paraphrases Boutyline’s definitions in ways that at times create issues with clarity. Many of these issues arise because, unlike Boutyline, the author does not use the conventional distinction between “observed” and “manifest,” instead using a variety of terms like “overt” or “seeming” to mean “observed” (or “manifest”), and terms like “logics” to mean “latent.” I strongly urge the author to use the observed/latent distinction because it makes the model much clearer.

For example, in the model, individuals in the same schematic class are theorized to share the same latent schema, and their sets observed responses are the product of transformations applied to these schemas. The author tries to get this distinction via statements like “Persons in the same schematic class may provide seemingly different (or even opposed) responses.” This phrasing is misleading because the responses are not just “seemingly” different: the observed responses are truly different, but different observed responses can come from one and the same latent schema.

Outside of this section, on p.8., the author writes that “different taste schemas will define distinct populations of omnivores and univores who stand opposed in terms of overt choices by virtue of sharing the same logic of taste.” This could also be improved by moving from talking about “overt choices” and “logic of taste” to the far clearer observed/manifest distinction. (But that sentence is still not clear to me even when I paraphrase it in my head: the main culprit here may be the phrase “who stand opposed,” which has no clear interpretation in this context.)

The phrase “This is done by assigning respondents to disjoint schematic classes based on a measure of similarity that is sensitive to the fact that two persons may belong to the same schematic class even though their vector of responses are (usually monotonic) transformations of one another (and thus overtly different)” could again be clarified via this distinction. This statement is also a bit off, because SCA is a

generic framework that doesn't assume anything about the characters of the transformations. Boutyline (Boutyline 2017:385) phrases this same idea as "Individuals' taste patterns are manifestations of [latent] schemas and are created from these schemas via one of a finite set of schematic transformations."

SUGGESTION: I strongly urge the author to stick more closely to the language used by Boutyline (2017), which is generally clearer than the author's paraphrasing. This is especially the case with the language of observed/manifest. The author should also address the unclear phrasings I pointed out. To save time, the author could consider using direct quotes.

SMALLER ISSUES:

7. The author initially uses the term "internal validity" correctly on the first line of page 4, but then overwhelmingly uses it incorrectly throughout the paper, esp. in (4.3) through (5.) Specifically, the author's validation studies use a behavioral outcome to evaluate schematic assignment. S/he calls this "internal" validation. But internal validity refers to just the inner workings of the study's/method's design: behavioral outcomes and/or demographic predictors thus both seem more external than internal. The tests the author is doing are more accurately described as "criterion validity," or perhaps simply "behavioral validation." On the other hand, the SEM goodness-of-fit test described in footnote 1 tests whether the method does what it claims to do, and is thus an internal validity test (the author implies otherwise.) That SEM test is also incorrectly cited: it is originally from Boutyline and Vaisey (2017).

8. The author claims that his or her validations "constitutes strong evidence" for the validity of CCA and EI. I believe the author should be more modest about the strength of this evidence. The study indeed provides a behavioral validation of the measures: but the behavioral outcome is self-reported rather than observed; and is also measured at the same time as the tastes via similarly phrased survey items, which could induce a consistency bias and other instrument-driven correlations. I would thus recommend that the author avoid calling this evidence "strong." The author also states that these results constitute strong evidence for a "spreading activation interpretation" of CCA results. A related claim is also made in the discussion ("if we read the schema correlation as a possible conduit for 'spreading activation') The analyses in this paper do show that this framework may be a useful methodological tool studying schemas (and one I personally hope other sociologists consider adopting.) But as an actual causal diagram of where tastes come from, it is certainly a case of "all models are false but some models are useful," since change in tastes is very likely exogenous. I thus urge the author to stick with a language that distinguishes the methodological usefulness of a model and its suitability as an interpretation of these schemas.

9. The claim "I showed that we can leverage techniques developed for the analysis of correlation networks to go beyond eyeballing" (p. 9) seems to suggest that this is an innovation: but Goldberg (2011) already examined schema networks via the clustering coefficient, and many others have done so in other contexts.

10. The paper's discussion seems to be overwhelmingly dedicated to the validation study, whereas the abstract and front end mostly dealt with the replication and the question of behavioral stability. The author may wish to adjust the discussion to make the paper feel more coherent.

11. On p.7 (and related figures): For consistency with "Anything (but) Country," the class "Anything (but Country and Rap)" should probably be called "Anything (but) Country or Rap." The implication of this parenthesis placement is that the class combines those who don't listen to country or rap ("Anything but Country or Rap") with those who listen to only one or both of those genres ("Anything Country or Rap"). Relatedly, "Hip/Hop Rap" should be "Hip Hop/Rap." Also, the author generally uses the term "hiphop," in the figures, but in the text s/he almost always calls it "rap." This should be made consistent.

12. Figures 3, 5 and 7, and to a lesser extent 2, are very hard to read due to the miniscule font sizes. The fonts need to be far larger to be legible. The figures themselves should probably be larger too.

1. The manuscript contains a number of typos.

- P.6: The formula for EI has an extra = sign after the summation symbol
- P.6: “Subjecting the data to CCA, yields” should probably not have that comma
- In first line of section 4.2, “aren” should be “are.”
- Typo on first line of p. 7: P.7: “as figure 3a, shows” shouldn’t have that comma
- P.7: “Figure 3d. Which” should probably be “Figure 3d, which”
- P.7: “unlikely Boutyline” should be “unlike Boutyline”
- P.4: Sentence starting with “Goldberg (2011) used internal” has many errors. Copy-and-paste?